

KAUSHIK BORA

Borlaug Institute for South Asia (BISA-CIMMYT), NASC Complex, New Delhi – 110012, India

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PROFESSIONAL PROFILE

Researcher in the field of agricultural and development economics. Experienced in climate and policy research related to climatic impacts, adaptation, and policy analysis. Skilled in econometric applications, meta-analysis, and data analysis, with several years of project management and partnership experience with governmental and development-focused institutions. Experience in collaborating with institutions of South Asia in the agriculture sector and extensive work focused on several parts of India.

RESEARCH INTERESTS

Agricultural and Development Economics · Environmental Economics · Climate Change Economics

PROFESSIONAL EXPERIENCE

Agricultural Economist (Research Associate)

June 2022 – Present

Borlaug Institute for South Asia (BISA-CIMMYT), New Delhi

- Involved with the *Atlas of Climatic Adaptation in South Asian Agriculture (ACASA)* and *Next Generation Bundled Crop Insurance in South Asia*.
- Use of high-resolution gridded data for climatic risk assessment.
- Application of spatial econometric models for adaptation effectiveness.
- Leading a team of socio-economists on the suitability and scalability of adaptation options.
- Mixed methods research for stakeholder prioritisation and gender-responsive scaling.
- Imparting training on climate risk management, meta-analysis and responsible scaling in agriculture.
- Development of a climate-smart agriculture index for policy prioritisation.

Visiting Scientist

October – November 2024

Evans School of Policy Analysis and Research, University of Washington, Seattle

- Work on cost-benefits and gender suitability of adaptation options.
- Application of spatial econometric models for adaptation effectiveness.

Project Manager, Climate Change

January – May 2022

MS Swaminathan Research Foundation, Chennai

- Led monitoring of projects related to climate change.
- Impact assessment of climate change in agriculture for major crops in Odisha and Tamil Nadu.
- Development of a composite index indicator for the climate change adaptation project.

Research Fellow

July 2017 – January 2022

Economic Analysis Unit (EAU), Indian Statistical Institute, Bangalore

- Doctoral research on the effect of climatic extremes on fertilizer use in Indian agriculture using panel econometric techniques.
- Capacity building of stakeholders and research collaborators in workshops.

Technical Assistant

September 2016 – April 2017

EDUCATION

- 2015–22 **PhD in Development Studies**
Indian Statistical Institute, Bangalore
(Degree awarded by Tata Institute of Social Sciences, Mumbai)
Thesis: A Study of Climatic Factors and Fertilizer Use in Indian Agriculture
- 2013–15 **M.Sc. Agricultural Economics**
Punjab Agricultural University, Ludhiana
Thesis: Productivity Analysis of Rice and Wheat in Indo-Gangetic Plains of India
- 2009–13 **B.Sc. in Agriculture**
Assam Agricultural University, Jorhat

PUBLICATIONS

1. Aggarwal, P., Shirsath, P., Gangopadhyay, P., Rahman, H., Rao, C.A.R., Ojha, R.B., Rathnayake, U.K., Barik, A., Deo, A., **Bora, K.**, Bandyopadhyay, S., Singh, S., Goswami, U.P., Pragya, P., Gupta, R. (2026). A web-based, gridded Atlas of multi-hazard, multi-species climate risks and technology options to support adaptation planning in South Asian agriculture. *Climate Services*, 42, 100671. <https://doi.org/10.1016/j.cliser.2026.100671> (Elsevier)
2. Maiti, S., Panja, A., Deo, A., **Bora, K.**, Garai, S., Shirsath, P., and Aggarwal, P.K. (2026). Climate adaptation in South Asian livestock production systems: Emerging trends and contextual insights. *Tropical Animal Health and Production*, 58(4), 233. <https://doi.org/10.1007/s11250-026-05030-9> (Springer Nature)
3. Gupta, R., **Bora, K.**, Gangopadhyay, P., and Aggarwal, P.K. (2026). Who adopts what? A systematic global review of gender dynamics in climate-smart agriculture. *Agricultural Systems*, 234, 104688. <https://doi.org/10.1016/j.agsy.2026.104688> (Elsevier)
4. **Bora, K.**, Shirsath, P.B., and Aggarwal, P. (2025). Hotspots of climate smart agriculture and climatic risks in India. *Regional Environmental Change*, 121(25). <https://doi.org/10.1007/s10113-025-02456-z> (Springer Nature)
5. Jain, N., Verma, H., Deo, A., **Bora, K.** et al. (2025). Effect of climate change adaptation options on maize yield across different agro-climatic zones in South Asia: A meta-analysis. *Agronomy for Sustainable Development*, 45(78). <https://doi.org/10.1007/s13593-025-01075-6> (Springer Nature)
6. Kumari, V.V., Deo, A., **Bora, K.**, Rao, C.R., Raju, B.M.K., Chandran, M.S. et al. (2025). Adapting sorghum and other millets to climate challenges: An integrated bibliometric and meta-analysis of global literature. *Journal of Agriculture and Food Research*, 21, 101987. <https://doi.org/10.1016/j.jafr.2025.101987> (Elsevier)
7. Ansari, M.A., Joshi, H., Mehta, P., Deo, A., **Bora, K.**, Rao, C.R. et al. (2025). Climate-smart and risk-resilient adaptation strategies for potato production: A meta-analysis for South Asia. *Environmental Challenges*, 101183. <https://doi.org/10.1016/j.envc.2025.101183> (Elsevier)
8. **Bora, K.** (2022). Rainfall shocks and fertilizer use: A district level study of India. *Environment and Development Economics*, 1–22. <https://doi.org/10.1017/S1355770X21000413> (Cambridge University Press)
9. **Bora, K.** (2022). Spatial patterns of fertilizer use and imbalances: Evidence from rice cultivation in India. *Environmental Challenges*, 27, 100452. <https://doi.org/10.1016/j.envc.2022.100452> (Elsevier)

10. **Bora, K.**, Bhattacharya, S., and Niyati, S. (2018). Paddy prices and marketing agencies in India: An empirical analysis of effectiveness of public procurement. *Agricultural Economics Research Review*, 31(Conf. Spl.), 157–166. 10.22004/ag.econ.281153

WORKING PAPERS

1. **Bora, K.**, Gupta, R., Gangopadhyay, P., Rama Rao, C.A., Raju, B.M.K., Sapkota, B., and Aggarwal, P.K. Gender responsive technology assessment index: Scaling inclusive climate smart interventions in data-limited settings. (Under Review). [Preprint Link]
2. Maiti, S., **Bora, K.**, Zade, S., Yadav, B., Sanjyal, S., Shirsath, P., Garai, S., Runa, R.A., Weerasinghe, W.M.P.B., Thornton, P., and Aggarwal, P.K. Identifying climatic risk hotspots for the dairy production system of South Asia: An integrated analysis with projections for 2050s and 2080s. (Under Review). [Preprint Link]
3. **Bora, K.** and Rahut, D. How farmers adjust irrigation during climate extremes? Insights from rice farming in India. (Under Review).
4. **Bora, K.**, Mkondiwa, M., and Aggarwal, P.K. Targeted strategies for closing yield gaps to achieve food self-sufficiency under current and future demand scenarios: Evidence from India. (Under Review).
5. **Bora, K.** and Aggarwal, P.K. Integrating yield gaps and sustainable intensification for a spatially explicit prioritization framework in agriculture. (Under Review).
6. **Bora, K.** Dynamic adjustments in fertilizer use under weather shocks: Micro-level evidence from India. (Under Review).
7. **Bora, K.** and Niyati, S. Trends in female labour absorption in rice cultivation: Evidence from India. Paper presented at the 31st Virtual Conference of the International Conference of Agricultural Economists. [Preprint Link]

SEMINAR AND CONFERENCE PRESENTATIONS

2025 Gender Responsive Technology Assessment Index: Scaling Inclusive Climate Smart Interventions in Data-Limited Settings. ANAPRI–World Bank Stakeholder Conference on “Leveraging the Agri Food System for Jobs”, Kigali, Rwanda, November 3, 2025.

2024 Yield Gap Decomposition and Technical Inefficiencies Under Climatic Risks in Monsoon Rice (poster). 32nd International Conference of Agricultural Economists (ICAE), August 2–7.

2021 Spatial Patterns of Fertilizer Use and Imbalances: Evidence from Rice Cultivation in India. ADBI Conference on Environmental Challenges and Agricultural Sustainability in Asia, December 8–10.

2021 Effects of Rainfall Shocks on Chemical Fertilizer Application: A Plot Level Study of Paddy in India. 15th Annual Meeting, Environment for Development (EfD) Initiative, November 15–19.

2021 Rainfall Shocks and Fertilizer Use: A District Level Study of India (poster). 6th International Climate Change Adaptation Conference, Adaptation Futures 2020, October 5–8.

2019 Fertilizer Use and Rainfall Shocks in Indian Agriculture. Tenth Biennial Conference, Indian Society of Ecological Economics, CESS, Hyderabad, November 6–8.

SOFTWARE KNOWLEDGE

Stata (Proficient) · R (Proficient) · Python (Beginner) · QGIS (Beginner)

TEACHING EXPERIENCE

- 2016 Teaching Assistant, Indian Statistical Institute, Bangalore
Course: Research Methodology for Development Studies
- 2021 Teaching Assistant, Indian Statistical Institute, Bangalore
Course: Poverty and Inequality
- 2022 Guest Faculty, Indian Statistical Institute, Bangalore
Course: Econometrics for Social Science
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